地脈感知

SeismicSense

The Last Defense



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Why Centralized Warnings Fail

The Epicenter Blind Zone

- **P-Wave Lag:** Regional systems rely on distant travel time.
- **Zero Lead Time:** Quakes directly underneath outrun cloud alerts
- **JMA Limitation:** Strong shaking arrives before remote warnings



Infrastructure Blackout

- **Infrastructure can fail:** Earthquakes can disrupt electricity, Wi-Fi, broadband, cellular networks, and routers.
- **Critical timing problem:** The system is most valuable when the infrastructure it depends on is most likely to be unavailable.
- **Need for resilience:** Emergency warnings should have dedicated, reliable communication paths that can continue operating when conventional networks fail.

Delayed access to guidance

- People need clear, immediate instructions when an emergency occurs.
- Searching for reliable safety information during a stressful situation can cost valuable time.

Our Solution: SeismicSense

How It Works:

1. **Monitor Motion**
   Each node uses an MPU6050 accelerometer and gyroscope to measure movement and send readings to its ESP32.

2. **Evaluate Unusual Motion**
   Each ESP32 analyzes sensor readings and checks whether configured motion criteria are met.

3. **Communicate Between Nodes**
   ESP32 nodes wirelessly share motion alerts using ESP-NOW, allowing the system to identify reports from multiple locations without internet access.

4. **Trigger Local Alerts**
   When detection criteria are met, the system activates a buzzer and LED to provide a local warning.

5. **Access Emergency Guidance**
   Users connect their phones to the gateway ESP32's local Wi-Fi network and open an emergency webpage hosted directly on the device.

6. **Keep Essential Information Available**
   Safety instructions and system status are stored or served locally, allowing the emergency webpage to work without external internet access.

- **SeismicSense** is A locally accessible, offline-ready emergency assistance system that combines motion monitoring, local alerts, and a phone-accessible emergency webpage to help people access essential safety information when connectivity is limited.



Chapter 1: Core Sensing Firmware

Direct Register Access: Polls 0x3B over Wire.h without library locks.

100-Sample Calibration: Isolates resting gravity vector at boot.

5-Sample Moving Average: Smooths high-frequency electrical noise at 50Hz.

NPN Transistor Driver: Amplifies GPIO 4 current for high-decibel siren.

$$M = \sqrt{a_x - B_x^2 + a_y - B_y^2 + a_z - B_z^2}$$

False alarm detection

STATE_IDLE

50Hz background structural polling.



STATE_VERIFYING

1000ms persistence filter blocks door slams.

STATE_EMERGENCY

Latches dual-tone siren for 15 seconds.

Chapter 2: On-Demand SoftAP and Captive Portal

📶 **Dynamic SoftAP Activation**
Network stays off in IDLE; spawns “SeismicSense” upon emergency.

📍 **Wildcard DNS Redirection**
Port 53 intercepts all queries to trigger OS captive portal popups.

🎛️ **Live Mobile Dashboard (192.168.4.1)**
Serves Drop/Cover/Hold guidance plus 500ms JSON telemetry updates.

Chapter 3: 6-Axis Accelerometer + Gyroscope Fusion

Sensor Channel	MPU6050 Register	Full-Scale Range	Trigger Threshold	System Role
3-Axis Accelerometer	0x3B (6 Bytes)	±4g (8192 LSB/g)	**1.50 m/s²**	Primary Linear Trigger
3-Axis Gyroscope	0x43 (6 Bytes)	±500°/s (65.5 LSB)	**15.0 °/s**	Rotational Confirmation
Hardware DLPF	0x1A (Config)	21Hz Bandwidth	**700ms Fast-Lock**	High-Confidence Boost

Distributed ESP-NOW Structural Consensus

0 NODES ACTIVE

Normal Monitoring

200ms ESP-NOW heartbeats on Channel 1.

1 NODE ACTIVE

Localized Bump

Single-node disturbance isolated during verification.

2+ NODES AGREE

Global Emergency

Synchronized building-wide sirens and portal launch.

Verified System Performance

50Hz

6-Axis Sampling Rate

700ms

6-Axis Verification Time

3 Nodes

ESP-NOW Mesh Array

100%

Offline Independence

Hardware

ESP32 Microcontroller:

- The brain of the system. It processes sensor readings, detects unusual movement, and controls the alarm and local Wi-Fi webpage.

MPU6050 Motion Sensor:

- Measures acceleration and rotational movement to help identify unusual motion.

NPN Transistor:

- Acts as an electronic switch, allowing the ESP32 to control the higher-current siren.

Buzzer:

- Produces a loud audible warning when the system detects movement that meets its emergency criteria.

LED Indicator:

- Provides a visual alert when the system is triggered.

Connecting Wires & Power Supply:

- Connect the components and provide the power needed for operation.

How they work together:

The MPU6050 measures movement → the ESP32 processes the data → the LED and siren provide alerts.

GitHub Repository

- https://github.com/alextma-netizen/Seismic-Sense

“Life happens wherever you are ,
Whether you make it or not.”



THANKS